

Data Cleaning Walkthrough

SME Capital Matching, Funding Readiness project

This document records the preparation of the capital funding application dataset: **1,186 applications** submitted by small and medium-sized enterprises across South Africa through an online form between May 2025 and June 2026. The source is a single CSV file, `Datasets/capital_matching_applications.csv`, of 1,186 rows and 27 columns, one row for each application and one column for each question posed by the form. All monetary values are stated in rand (R).

Approximately one response in ten was received in an unusable state: capital funding requests padded with underscores; 2023 and 2024 revenue recorded as worded ranges rather than figures; telephone and registration numbers recorded in columns intended for revenue and shareholder counts; B-BBEE ratings recorded as text; ownership percentages expressed using two different dash characters; one province recorded under two names; unreadable symbols within the company descriptions; and the same business submitted on more than one occasion. Certain of these defects prevent a calculation from completing. Others present as valid responses and materially distort the outcome without detection.

Each defect is addressed below: how it presented, the remedy applied, and the limitations of that remedy. The scope is the cleaning stage only. The output is `Datasets/Capital_Matching_Cleaned_Data.xlsx`, a register of 1,116 unique businesses across 41 columns. The scoring work that reads that file, which assesses each business for funding readiness and assigns applicants to tiers, is documented separately.

Source code. Cleaning: [`Data\_Cleaning.ipynb`](#). Scoring: [`Funding\_Readiness\_Segmentation.ipynb`](#).

Confidentiality. The original CSV file contains actual names, email addresses, telephone numbers and registration numbers. None are reproduced in this document. Every example shown is either fabricated or a description of a pattern.

Contents

1. The dataset
2. The condition of the source data
3. Capital funding requests and 2025 revenue figures
4. Responses entered in the wrong column
5. Company descriptions that cannot be read
6. Standardisation of the fixed-choice responses
7. The two rows holding values other than ownership percentages
8. Removal of duplicate applications
9. The dataset before and after cleaning

1. The dataset

The source file comprises **1,186 rows and 27 columns**. The table below lists all 27, grouped by purpose, together with the short name used throughout this document and the additional columns introduced during cleaning. No cleaned value is written over an original response: each is placed in a new column alongside the original, so that the two remain available for comparison.

Group	The 27 columns as received	Columns added during cleaning
Identity and location	company_name , company_reg_no , industry , province , city_town , area , head_office_address , company_overview	none
Capital funding request	funding_ask_raw , funding_type , funding_purpose	funding_ask_zar
Revenue	revenue_band_2023 , revenue_band_2024 , revenue_2025_raw	revenue_scale_2023 , revenue_mid_2023 , revenue_scale_2024 , revenue_mid_2024 , revenue_2025_zar
Size and jobs	employees_raw , jobs_anticipated_raw , shareholders_raw	employees , jobs_anticipated , shareholders
Ownership	black_ownership_band , women_ownership_band , youth_ownership_band , disability_ownership_band , bee_level_raw	black_ownership_pct , women_ownership_pct , youth_ownership_pct , disability_ownership_pct , bee_level
Contact and submission	applicant_name , contact_name , email , contact_number , created_at	none
	27 columns	14 columns

Three columns are referred to throughout: `funding_ask_raw` , the amount of capital funding requested; `revenue_2025_raw` , what the business earned in 2025; and `revenue_band_2023` and `revenue_band_2024` , what it earned in those two years, selected from a list of ranges rather than entered as a figure.

The suffix `_raw` denotes the response exactly as the applicant entered it, and the cleaned equivalent is held in the adjacent column: `funding_ask_raw` holds `600000_____` , while `funding_ask_zar` holds `600000` . Twenty-seven original columns plus fourteen additions gives the final total of 41.

2. The condition of the source data

Each of the 27 columns was reviewed in full. The two columns holding monetary values entered by the applicant, `funding_ask_raw` and `revenue_2025_raw` , contained digits embedded in underscore padding (`600000_____` , `750_000`), together with a small number of figures stated in billions and trillions. The 2023 and 2024 revenue columns contained worded ranges (`0 to R500K` , `Above R50M`) in place of figures. The shareholder count column contained personal names, telephone numbers and complete sentences. The four ownership percentage columns contained ranges written with two different dash characters (`50-74%` , `50~74%`). Registration numbers were recorded in inconsistent formats, B-BBEE ratings as text (`Level 1`), and company descriptions carried symbols that have no meaning in an English sentence.

The column headings were likewise unusable, several carrying stray spacing and duplicated colons:

Heading as recorded by the form	Renamed to
Name:: Name & Surname	applicant_name
Funding Ask (Amount):	funding_ask_raw
Increase of jobs anticipated through the capital access	jobs_anticipated_raw
What percentage of your business is owned by women? (Please select one)	women_ownership_band

Such headings are cumbersome to reference and readily mistyped, and a single mistyped character results in the wrong column being processed. All 27 were renamed to the short names given in section 1 before any further work commenced.

The file was loaded as plain text. Python's default behaviour on opening a CSV file is to determine the type of data each column holds and convert the contents accordingly, which in this instance would have caused damage before any inspection was possible. The clearest illustration is a leading zero: registration and contact numbers both begin with one, and as soon as either is treated as a number that zero is discarded, with no subsequent means of establishing whether it was ever present. The same exposure applies to the two monetary columns. Loading as text preserves every character exactly as recorded.

```
raw = pd.read_csv(RAW_FILE, dtype=str)
```

3. Capital funding requests and 2025 revenue figures

Almost every capital funding request and 2025 revenue figure was padded with underscores, in the form `600000_____`. This was the most extensive defect in the file. The number of records affected is high, but the fault is consistent, and the remedy is therefore straightforward. The 2023 and 2024 revenue columns are not affected, having been recorded as worded ranges, and are addressed in section 6.

A small number of entries were malformed in other respects:

Nature of the defect	Example	Records affected
Underscores appended to the figure	<code>600000_____</code>	1,176
Underscores within the figure	<code>750_000, 250000_00</code>	a small number
Commas, spaces or a leading R	<code>R8, 461, 029.00</code>	3
Capital funding request stated in words	<code>R50million</code>	1
Funding request or revenue that cannot be genuine	<code>1000000000000000</code>	11, across both columns

The remedy applied: remove every character that is not a digit or a full stop, and where two full stops remain, remove the first, since the final one is the decimal point.

```
def parse_money(x):
    if pd.isna(x):
        return np.nan
    s = re.sub(r"^[0-9.]", "", str(x)) # keep only digits and full stops
    s = re.sub(r"\.(?=.*\.)", "", s) # if two full stops, drop the first
    if s in ("", "."):
        return np.nan
    return float(s)
```

Because the fault is almost entirely a single repeated pattern, one rule resolves the majority of cases in a single pass, leaving the underlying figure unaltered: `600000_____` becomes `600000`, and `R8, 461, 029.00` becomes `8461029.00`.

Two limitations

The rule cannot correct a figure entered incorrectly at source. A number of entries read as figures such as `1000000000000000`, being one hundred trillion rand. These are genuine digits with no padding surrounding them, and cleaning offers no remedy because there is nothing to remove. A threshold was therefore applied after cleaning: any capital funding request or 2025 revenue figure above **R2 billion** is treated as a data entry error and cleared, removing **11** entries across the two columns. The threshold sits materially above any figure a small or medium-sized enterprise would credibly report or request, so no genuine value is captured by it.

A cleared entry is left blank rather than converted to zero. That distinction is material at the scoring stage: a blank 2025 revenue figure is treated as unknown, not as a business reporting no income, and neither advantages nor disadvantages the business.

The rule handles one entry incorrectly. One applicant entered `R50million` in place of a figure. Because the rule removes all letters, the entry resolves to `50`, recording the business as requesting fifty rand rather than fifty million. This single record was corrected individually, to `5000000`.

4. Responses entered in the wrong column

Certain responses were entered in the wrong place on the form: telephone numbers in the 2025 revenue column, and company registration numbers in the shareholder count column. This is more difficult to detect than the preceding defect, which is what makes it the greater risk. A telephone number consists only of digits, so it passes through the rule described in section 3 unaltered and is indistinguishable from a genuine figure. A mobile number left in the 2025 revenue column loses its leading zero and becomes 821,234,567, recording the business as having earned R821 million in 2025. It would rise to the top of the revenue rankings and inflate every revenue average. A single missed entry is capable of changing the conclusions of the entire exercise.

Where a response plainly did not answer the question posed it was therefore cleared rather than retained. Clearing it costs one data point; retaining it costs the credibility of the result.

The clearest instance is `shareholders_raw`. Of the 1,179 responses present, **72** were not numbers at all, approximately 6% of the column: mostly personal names, in some cases a list of directors, and otherwise telephone numbers, company names or lists of client organisations. A small number gave the correct answer in words, `one`, `Myself`, `sole`, and these are read as `1`. All other non-numeric responses are discarded and left blank.

```

WORDED_COUNT = {"one": 1.0, "myself": 1.0, "sole": 1.0, "sole owner": 1.0}

def parse_count(x):
    if pd.isna(x):
        return np.nan
    s = str(x).strip()
    if s.lower() in WORDED_COUNT:
        return WORDED_COUNT[s.lower()]
    return pd.to_numeric(s, errors="coerce")

```

5. Company descriptions that cannot be read

`company_overview` holds paragraphs of narrative text. Where a description should have held -, é or ', the source file held â or Ã@ instead, leaving a stray character in the middle of a word.

The column also held characters that occupy a position in the text but display nothing at all. These most probably transferred with text composed in a Word document, a PDF or an email and then pasted into the form. Certain positions in the character set were designated long ago as instructions to machines rather than for reading, and others were reserved and never allocated, so there is nothing for a screen to display. Tabs and line breaks belong to the same group but hold the paragraph structure together, and were therefore retained.

Such characters cannot be identified by reading the column, since they display nothing on screen or at most a gap resembling an ordinary space. They were identified instead by checking every character against the positions designated for machine instructions. Excel cannot store them, and writing the cleaned data out to `Capital_Matching_Cleaned_Data.xlsx` fails if even one remains present.

```

def clean_cell(v):
    if not isinstance(v, str):
        return v
    return "".join(
        ch for ch in v
        if ch in "\t\n\r" or (ord(ch) >= 0x20 and ord(ch) != 0x7f
            and not (0x80 <= ord(ch) <= 0x9f)) # remove the invisible ones
    )

```

6. Standardisation of the fixed-choice responses

Provinces. The province column holds nine different spellings. **36 applications** recorded `KZN` where the remainder of the file records `KwaZulu-Natal`. To a computer these are simply two different words, and left uncorrected those 36 businesses would have been counted as a tenth province. Every spelling was mapped to a single standard version.

```

def fix_province(x):
    return PROVINCE_FIX.get(str(x).strip().lower(), str(x).strip())
# "kzn", "kwazulu natal", "KwaZulu-Natal" -> "KwaZulu-Natal"

```

Ownership percentages. Four columns record Black, women, youth and disability ownership. Some responses use a short dash - and others a long dash -. The two appear almost identical but are treated

as distinct, so 50-74% and 50-74% would have been recorded as two different responses despite carrying the same meaning. The dash characters were standardised and each range converted to a single number drawn from within it, so that ownership could be compared and scored.

```
PERCENT_BAND_MIDPOINT = {"0%":0, "1-24%":12.5, "25-49%":37,
                          "50-74%":62, "75-99%":87, "100%":100}

def parse_percent_band(x):
    return PERCENT_BAND_MIDPOINT.get(str(x).strip().replace("-", "-"), np.nan)
```

These are rounded values representing the range, not its exact centre. Their purpose is to place each response on a scale of 0 to 100 so that the four columns can be averaged and compared, not to state what any individual business's ownership split actually is.

Revenue ranges for 2023 and 2024. These two columns hold a range selected from a list of seven, from 0 to R500K up to Above R50M. A range cannot be used in a calculation, so each was assigned two values: a rank from 0 to 6, which conveys size and year-on-year movement, and a rand figure drawn from the middle of the range, which permits the funding request to be compared against revenue. R1 000 001 to R5M becomes rank 2 and R3 million. The rand figure is an estimate, since every business within a range is assigned the same value and Above R50M has no upper limit, and it therefore serves for grouping and comparison rather than as a statement of earnings.

The rank exists so that the two years can be subtracted, the 2024 rank less the 2023 rank giving the movement, which the scoring work uses directly, capped at two ranges in either direction. Subtracting the rand figures instead would distort the result, because the intervals are uneven: R250,000 to R750,000 is a step of half a million, while R35 million to R75 million is a step of forty million. Ranking gives every step equal weight.

```
REVENUE_BAND_ORDER = {"0 to R500K":0, "R501K to R1M":1, "R1 000 001 to R5M":2,
                      "R5 000 001 to R10M":3, "R10 000 001 to R20M":4, "R20 000 001 to R50M":5,
                      "Above R50M":6}

REVENUE_BAND_MIDPOINT = {"0 to R500K":250_000, "R501K to R1M":750_000,
                         "R1 000 001 to R5M":3_000_000, "R5 000 001 to R10M":7_500_000,
                         "R10 000 001 to R20M":15_000_000, "R20 000 001 to R50M":35_000_000,
                         "Above R50M":75_000_000}

for yr in ["2023", "2024"]:
    band = f"revenue_band_{yr}"
    df[f"revenue_scale_{yr}"] = df[band].map(REVENUE_BAND_ORDER)      # rank 0-6
    df[f"revenue_mid_{yr}"] = df[band].map(REVENUE_BAND_MIDPOINT)    # rand figure
```

B-BBEE rating. B-BBEE is South Africa's broad-based black economic empowerment rating, running from Level 1 to Level 8, with Level 1 the strongest. The scale runs in the opposite direction to expectation, so a lower number denotes a stronger position. bee_level_raw held the rating as a phrase, Level 1, Level 4, Level 8, which cannot be sorted, compared or scored. Every entry carried the word Level, so the column contained no usable numbers at all. Each was converted to the number it states.

```
def parse_bee_level(x):
    if pd.isna(x):
        return np.nan
    m = re.search(r"(\d+)", str(x)) # find the digits in "Level 4"
    return int(m.group(1)) if m else np.nan
```

7. The two rows holding values other than ownership percentages

The four ownership columns offered six responses to select from: 0%, 1–24%, 25–49%, 50–74%, 75–99% and 100%. All 1,186 rows were tested against those six, and exactly two failed. In those two the ownership columns hold responses relating to age, gender and race instead: 45 to 55 where a percentage belongs, Male in the women ownership column, Black in the Black ownership column.

Both were submitted on 29 May 2025 and are the two earliest applications in the file, and every application after them holds an ownership percentage. The form was evidently amended early in the collection period, which provides the assurance that the issue is closed rather than ongoing. No special handling was required: the conversion described in section 6 accepts a response only where it matches one of the six valid values, so these are returned as blank. No substitute value was entered, because there is no percentage that Male can reasonably be taken to mean.

8. Removal of duplicate applications

Certain businesses appear in the source file more than once. Duplicates were removed in two passes, retaining the first instance in each case:

```
# Pass 1: drop rows identical in every single field
df = df.drop_duplicates()
# Pass 2: drop rows with the same company, email and submission time
df = df.drop_duplicates(subset=["company_name", "email", "created_at"], keep="first")
```

The first pass identifies rows that are identical across all 27 columns, being the same application submitted twice, most probably the result of the submit button being pressed a second time. This removed 68 rows. The second pass is narrower, identifying rows that hold the same company name, email address and submission time but differ elsewhere, and removed a further 2. Together the two passes reduced the dataset from 1,186 rows to **1,116 unique business applications**.

A limitation applies here also. Duplicates are identified only where the company name and email address match exactly. Where the same business applied twice using two email addresses, or recorded its own name differently on the second occasion by adding (Pty) Ltd or omitting a space, the duplicate remains in the data. There is no dependable means of identifying those cases without risking the merger of two genuinely different businesses with similar names.

9. The dataset before and after cleaning

Measure	Before	After
Rows	1,186	1,116 unique businesses
Capital funding requests readable as numbers	7	1,109
2025 revenue figures readable as numbers	5	1,112
Shareholder counts holding words rather than numbers	72	0
Funding requests or revenue figures above R2 billion	11	0
B-BBEE ratings readable as a number	0	1,116
Applications recording the province as "KZN"	36	0

Approximately one response in ten was received in an unusable state. In the columns on which the scoring depends that figure is now below one in a hundred, each measure above holding a usable value for at least 1,109 of the 1,116 businesses. Anything still absent is recorded as unknown, not as zero.

The increase from 27 columns to 41 is deliberate. No original response was overwritten: each cleaned value is held in a new column alongside the raw response from which it was derived, so `funding_ask_raw` still holds `600000_____` while `funding_ask_zar` holds `600000`. Any reviewer is able to compare what was entered against what was used, and to question any decision recorded in this document.

`Datasets/Capital_Matching_Cleaned_Data.xlsx`, a single table of **1,116 businesses across 41 columns**, is the sole input to the scoring work.

SME Capital Matching, Funding Readiness project. All example values shown are fabricated or illustrative. No actual applicant data is reproduced.